

A WEIGHTED FEATURE REDUCTION METHOD FOR POWER SPECTRA OF RADAR HRRPS¹

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Abstract Feature reduction is a key process in pattern recognition. This paper deals with the feature reduction methods for a time-shift invariant feature, power spectrum, in Radar Automatic Target Recognition (RATR) using High-Resolution Range Profiles (HRRPs). Several existing feature reduction methods in pattern recognition are analyzed, and a weighted feature reduction method based on Fisher's Discriminant Ratio (FDR) is proposed in this paper. According to the characteristics of radar HRRP target recognition, this proposed method searches the optimal weight vector for power spectra of HRRPs by means of an iterative algorithm, and thus reduces feature dimensionality. Compared with the method of using raw power spectra and some existing feature reduction methods, the weighted feature reduction method can not only reduce feature dimensionality, but also improve recognition performance with low computation complexity. In the recognition experiments based on measured data, the proposed method is robust to different test data and achieves good recognition results.

Key words Radar Automatic Target Recognition (RATR); High-Resolution Range Profile (HRRP); Power spectrum; Feature reduction; Fisher's Discriminant Ratio (FDR)

I. Introduction

Statistical pattern recognition system can be subdivided into four phases, namely, data acquisition, data preprocessing, feature reduction (including feature extraction and feature selection) and classification. Usually raw features obtained in the first two phases are in a higher dimensional feature space. Feature reduction is a process of reducing dimension for raw features, which consists of feature extraction and selection. Feature extraction is to create new features based on transformations or combinations of raw features. Feature selection is to select the most useful subset of raw features. Thereby feature reduction is a key process to delete redundancy and decrease computation. Feature reduction methods can be divided into two classes. One is based on the minimum Mean Square Error (MSE) approximation, which leads to the minimum energy loss in the lower dimensional subspace, such as Principal Component Analysis (PCA) or Karhunen-Loève Transform (KLT), Singular Value Decomposition (SVD), Independent Component Analysis (ICA) and so on. The other is based on class separability measures, which leads to the maximum class separability in the lower dimensional subspace, such as Fisher's linear dis-

criminant, the minimum Bhattacharyya distance, auto-associative networks and so on.

High-Resolution Range Profile (HRRP) is the projection of sub-echoes onto the radar Line-Of-Sight (LOS), which are returned from the scatterers of target captured by wide-band radar signal. HRRP provides the scatterer distribution in the distance direction, which constitutes the significant characteristics of target structure. Thereby radar HRRP target recognition has received intensive attention from the Radar Automatic Target Recognition (RATR) community^[1-7]. However, HRRPs are sensitive to target-aspect, amplitude-scale and time-shift^[1-7]. The sensitivity of time-shift is caused by the move of range image in range window. The image is a section of vector extracted from part of the echo data by range window which contains the target and some remaining part. Even for a small time-shift of the range window, obvious change of image vector can take place. Therefore, all the traditional radar HRRP target recognition methods need time-shift search. In this way even using the fast correlation algorithm based on the Fast Fourier Transform (FFT) or Hilbert transformation, computation of classifier is also very large. If a time-shift invariant feature of HRRP can be obtained, not only the computation can be reduced, but also advanced classifiers can be used to improve recognition performance. As a time-shift invariant feature, higher-order spectra have been well studied by RATR community^[3,5-7]. The reference^[7] showed that power spectrum (the first order spectrum) has the best recognition performance among the higher-order spectra.

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This paper deals with the feature reduction methods for the power spectra of HRRPs. Several existing feature reduction methods are analyzed, and a weighted feature reduction method based on Fisher's Discriminant Ratio (FDR) is proposed. The proposed weighted feature reduction method uses the FDR vector of target power spectra to iteratively search for the optimal weight vector, and reduces feature dimensionality in accordance with the optimal weight vector. Compared with the method of using the raw power spectra and some existing feature reduction methods, the proposed method not only is robust to different test data, but also improves the recognition performance with low computation complexity in the recognition experiments based on measured data.

II. Some Existing Feature Reduction Methods

Fisher's linear discriminant is a class separability measure, which has two forms, FDR^[8,9] and Fisher's optimal projection subspace^[5,6,8]. FDR is applied to independent feature components, and Fisher's optimal projection subspace is applied to correlated feature components.

1. Direct feature reduction methods based on FDR

Assume that there are c targets, and let $\{\mathbf{x}_{ik} | k=1,2,\dots,K\}$ represent all N -dimensional features of target ω_i ($i=1,2,\dots,c$), where k denotes the sample number. FDR of the n -th feature component is defined as

$$F(n) = \sum_{i=1}^c \sum_{j \neq i}^c \frac{(\mu_i(n) - \mu_j(n))^2}{\sigma_i^2(n) + \sigma_j^2(n)}, \quad n=1,2,\dots,N \quad (1)$$

where $\mu_i(n)$ and $\sigma_i^2(n)$ represent the mean and the variance of the n -th feature component of target ω_i , respectively.

$$\mu_i(n) = \frac{1}{K} \sum_{k=1}^K x_{ik}(n) \quad (2)$$

$$\sigma_i^2(n) = \frac{1}{K-1} \sum_{k=1}^K (x_{ik}(n) - \mu_i(n))^2 \quad (3)$$

where, $n=1,2,\dots,N$, $i=1,2,\dots,c$.

The weights can be defined as

$$w(n) = \begin{cases} 1, & n_s = n, \quad \frac{F(n)}{\max_n \{F(n)\}} \geq \text{Th} \\ 0, & n_u = n, \quad \frac{F(n)}{\max_n \{F(n)\}} < \text{Th} \end{cases} \quad (4)$$

where $n=1,2,\dots,N$, and Th denotes the threshold.

The corresponding reduced feature vector is

$$\begin{aligned} \chi_{ik}(r) &= w(n_s) x_{ik}(n_s), \\ r &= 1, 2, \dots, d, \quad i = 1, 2, \dots, c, \\ k &= 1, 2, \dots, K \end{aligned} \quad (5)$$

where

$$d = L\{w(n) \neq 0 | n=1,2,\dots,N\} \quad (6)$$

d denotes the number of nonzero elements of weight vector $\mathbf{w}$. Thus the reduced feature vector is a d -dimensional vector.

2. Feature reduction method based on Fisher's optimal projection subspace

Fisher's optimal projection subspace is to seek a subspace $\Omega = \text{Span}(\mathbf{U}) \in R^{N \times d}$ to maximize the class separability, which can satisfy

$$\max_U J(\mathbf{U}) = \frac{\prod_{\text{diag}} \mathbf{U}^T \mathbf{S}_b \mathbf{U}}{\prod_{\text{diag}} \mathbf{U}^T \mathbf{S}_w \mathbf{U}} \quad (7)$$

where $\prod_{\text{diag}} \mathbf{A}$ denotes the product of all diagonal elements of the matrix $\mathbf{A}$, $\mathbf{S}_w$ and $\mathbf{S}_b$ represent the average within-class scatter matrix over all targets and the between-class scatter matrix, respectively.

$$\mathbf{S}_w = \sum_{i=1}^c P_i \mathbf{S}_i = \sum_{i=1}^c P_i \left[\sum_{k=1}^K (\mathbf{x}_{ik} - \mathbf{m}_i)(\mathbf{x}_{ik} - \mathbf{m}_i)^T \right] \quad (8)$$

$$\begin{aligned} \mathbf{S}_b &= \sum_{i=1}^c P_i (\mathbf{m}_i - \mathbf{m})(\mathbf{m}_i - \mathbf{m})^T \\ &= \sum_{i=1}^c P_i \left(\mathbf{m}_i - \sum_{i=1}^c P_i \mathbf{m}_i \right) \left(\mathbf{m}_i - \sum_{i=1}^c P_i \mathbf{m}_i \right)^T \end{aligned} \quad (9)$$

where $\mathbf{m}_i$ denotes the clustering center of target ω_i , $\mathbf{m}$ denotes the clustering center of all targets, and P_i denotes the *a priori* probability of target ω_i . By derivation^[5,6,8], this nonlinear optimization problem in Eq.(7) equals to a generalized eigenvalue decomposition problem of the matrix pencil $\{\mathbf{S}_b, \mathbf{S}_w\}$,

$$\mathbf{S}_b \mathbf{u}_j = \lambda_j \mathbf{S}_w \mathbf{u}_j, \quad j=1,2,\dots,d, \quad d < N \quad (10)$$

where $\mathbf{u}_j$, the j -th column vector of the matrix $\mathbf{U}$, is the eigenvector corresponding to the generalized eigenvalue λ_j ($\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d \geq \dots \geq \lambda_N$) of $\{\mathbf{S}_b, \mathbf{S}_w\}$. The corresponding d -dimensional feature vector is

$$\chi_{ik} = \mathbf{U}^T \mathbf{x}_{ik}, \quad i=1,2,\dots,c, \quad k=1,2,\dots,K \quad (11)$$

Thus the reduced feature vector is acquired from linear change of the raw feature.

III. A Weighted Feature Reduction Method Based on Fisher's Discriminant Ratio

1. Feature reduction in radar HRRP target recognition

Compared with ordinary pattern recognition,

several issues must be considered for radar HRRP target recognition.

Firstly, as mentioned in Section I, time-shift invariant features should be extracted as the raw features due to the time-shift sensitivity of HRRPs. Secondly, as mentioned in Section I, HRRPs are sensitive to target-aspect. According to the scattering center model, the variation of target-aspect will lead to different range shifts for different scattering centers on the target even within the aspect region where the scattering center structure remains unchanged, therefore, the HRRP, the amplitude of coherent sum of the complex returned echoes from scatterers in a range resolution cell, can be changed substantially. The time-shift invariant features (such as higher-order spectra, central moments and so on) of HRRPs are also sensitive to target-aspect variation. Thereby, the HRRPs' features of every target can be divided into many subsets according to different angular sectors, which are defined as feature vector sets of HRRP frames. Each feature vector set can be regarded as a vector stationary process, which equals to a pattern in pattern recognition^[3,5-7]. Although every target is in correspondence with many feature sets in radar HRRP target recognition, the recognition result mainly depends on comparison between the between-class nearest patterns. Therefore, when computing weight vector or optimal projection subspace by means of various feature reduction methods, we need to use the between-class nearest sample sets to calculate the related weight vector or optimal projection subspace, and then average them to get the final result. This is the point we need to pay attention to in the feature reduction of radar HRRP target recognition.

2. A weighted feature reduction algorithm based on FDR

The direct feature reduction method in Subsection II.1 is to select the feature components with larger FDRs to reduce feature dimensionality. In other words, the weights used in this method can only be 1 or 0. Considering that most classification algorithms use the similarity measure based on Euclidean distances, the Euclidean distances between samples will change, if raw feature components are multiplied by different weights. Therefore, a weighted feature reduction method based on FDR is proposed in this paper, in which feature components are multiplied by weights $w(n) \in [0,1]$ ($n = 1, 2, \dots, N$) to increase rationality of the contribution of feature components to Euclidean distance calculation. The optimal weight vector is sought by an it-

erative algorithm to make the FDRs of feature components converge. The algorithm is as follows.

(1) Initialization Let $p = 0$. Here the initialized feature vectors are the power spectra. Let $\chi_{ikl}^{(0)} = \mathbf{x}_{ikl}$ ($i = 1, 2, \dots, c$, $k = 1, 2, \dots, K$, $l = 1, 2, \dots, L$), where i denotes target class, k denotes the frame number and l denotes the feature vector number in a frame;

(2) Iterative process The between-class nearest feature sets of HRRP frames are searched among the clustering centers. Let

$$\{\chi_{ikl}^{(p)} \mid l = 1, 2, \dots, L\} \rightarrow \{\chi_{jm_j(i,k)l}^{(p)} \mid l = 1, 2, \dots, L\} \\ (i, j = 1, 2, \dots, c, i \neq j, k = 1, 2, \dots, K)$$

where the $m_j(i,k)$ -th frame of the j -th target is nearest to the k -th frame of the i -th target. Thus there are cK between-class nearest feature sets in all. If Eq.(1) is used to calculate the FDR vectors of all between-class nearest feature sets $\{\mathbf{F}_q \mid q = 1, 2, \dots, cK\}$, the average FDR vector is

$$\mathbf{F}_m = \frac{1}{cK} \sum_{q=1}^{cK} \mathbf{F}_q \quad (12)$$

Define the p -th iterative weight vector as

$$\mathbf{w}^{(p)} = \frac{\mathbf{F}_m}{\max_n \{F_m(n)\}} \quad (13)$$

(3) Stop condition If the weights of feature components converge, the iterative process ends. Here variance is used to measure the convergence. Let $\varepsilon > 0$, the variance of weight vector $\mathbf{w}^{(p)}$ is

$$\sigma_{\mathbf{w}^{(p)}}^2 = \frac{1}{N-1} \sum_{n=1}^N \left(w^{(p)}(n) - \frac{1}{N} \sum_{n=1}^N w^{(p)}(n) \right)^2 \quad (14)$$

If $\sigma_{\mathbf{w}^{(p)}}^2 > \varepsilon$, the $(p+1)$ -th iterative feature vector is

$$\chi_{ikl}^{(p+1)} = \mathbf{w}^{(p)} \bullet \chi_{ikl}^{(p)} \quad (15)$$

where “ $\bullet$ ” denotes dot production between vectors. Let $p = p+1$ and the iterative process continue. Otherwise, the weights converge, thus let the iterative process end;

(4) Feature reduction Let the optimal weight vector be

$$\mathbf{w}^* = \mathbf{w}^{(0)} \bullet \mathbf{w}^{(1)} \bullet \dots \bullet \mathbf{w}^{(p)} \quad (16)$$

The weights can be defined as

$$w(n) = \begin{cases} w^*(n), & n_s = n, \quad w^*(n) \geq \text{Th} \\ 0, & n_u = n, \quad w^*(n) < \text{Th} \end{cases} \quad (17)$$

where $n = 1, 2, \dots, N$. Substituting $\mathbf{w}^*$ into Eq.(5) gives the corresponding reduced feature vectors $\{\chi_{ikl} \mid i = 1, 2, \dots, c, k = 1, 2, \dots, K, l = 1, 2, \dots, L\}$.

A fact worth emphasizing is that the FDR remains unchanged whether the feature components

are weighted or not, according to Eq.(1). However, as mentioned above, the between-class nearest sets will change when the features are weighted, which leads to different FDR. This is one of the reasons that why the feature reduction method proposed in this paper has better recognition performance.

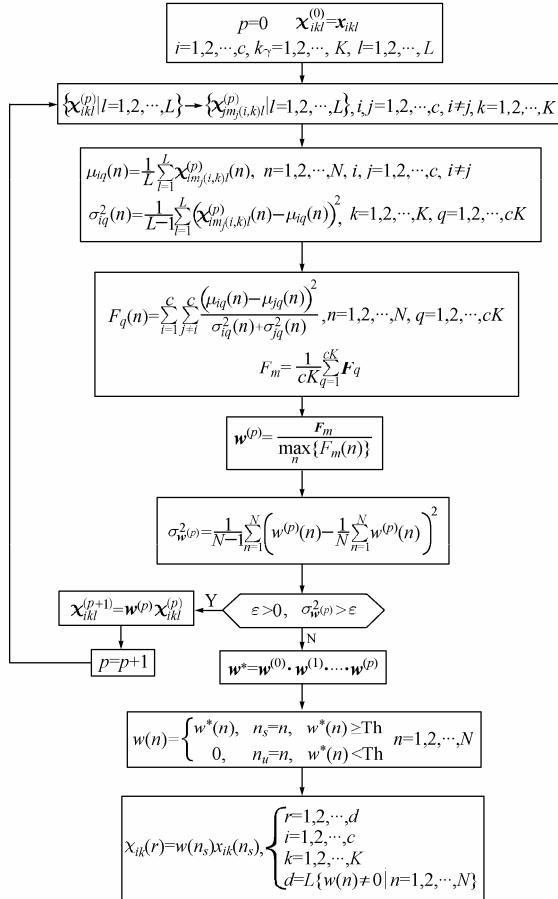


Fig.1 A weighted feature reduction algorithm based on FDR

IV. Recognition Experiments Based on Measured Data

1. The measured data

The results presented here are based on measured airplane data. The parameters of targets and radar are shown in Tab.1, and the projections of plane trajectories onto ground are shown in Fig.2, from which the aspect angle of airplane can be estimated according to its relative position to radar. As shown in Fig.2, the measured data are segmented. Training data and test data are from different data

segments, among which the 2nd and the 5th segments of Yark-42, the 5th and the 6th segments of An-26, the 6th and the 7th segments of Cessna citation S/II are taken as the training data, other data segments are taken as the test data. The training data cover almost all the target-aspect angles, but their elevation angles are different from those of the test data.

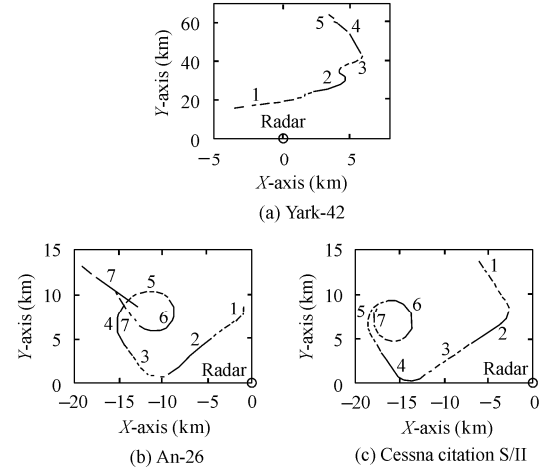


Fig.2 The projections of plane trajectories onto ground plane

2. Experimental results

When $p=3$, the weight vector $w^{(3)}$ converges in the experiment. Fig.3 shows the corresponding $w^{(0)} \sim w^{(3)}$ and the optimal weight vector w^* , and they include all of the weights corresponding to positive and negative frequency components of power spectra. Actually only a half of feature components and weights, namely, the ones corresponding to positive or negative frequency components, are applied to recognition. The weight vectors of the direct feature reduction method based on FDR and of the proposed weighted feature reduction method are shown in Figs.3(a) and 3(e), respectively. From the waveforms of weight vectors, we can see the feature components corresponding to lower frequency components have larger class separability than those corresponding to higher frequency components. For a comparison in recognition performance between the feature reduction methods, 18-dimensional feature vectors (reduced from the 128-dimensional power spectra) are used in the recognition experiments.

Tab.1 The parameters of planes and radar in the ISAR experiment

Radar		Planes								
Center frequency	Bandwidth	Yark-42			An-26			Cessna citation S/II		
		L (m)	W (m)	H (m)	L (m)	W (m)	H (m)	L (m)	W (m)	H (m)
5520 MHz	400 MHz	36.38	34.88	9.83	23.80	29.20	9.83	14.40	15.90	

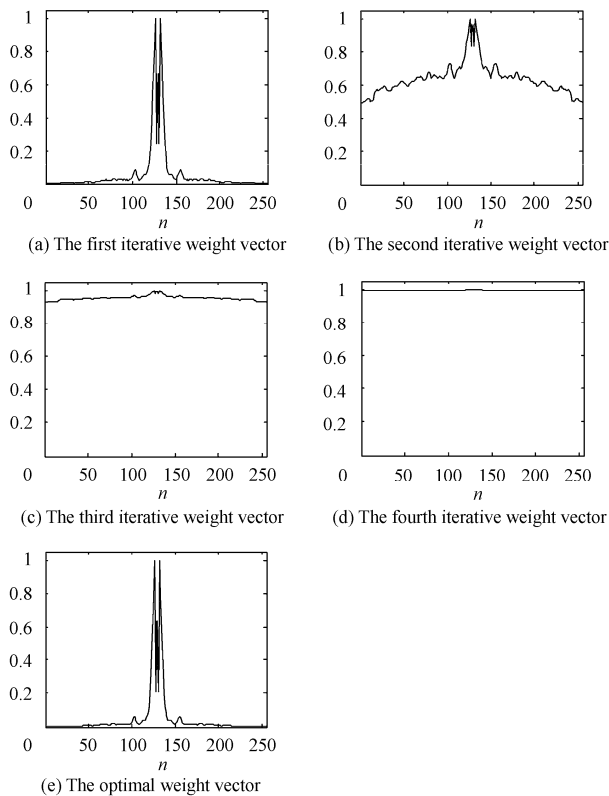


Fig.3 The iterative weight vectors and the optimal weight vector for power spectra of measured HRRPs using the iterative algorithm based on FDR

Template Matching Method under the Maximum Correlation Coefficient criterion (MCC-TMM)^[3-5,7] is used to compare the recognition performances of the feature reduction methods. The recognition rates of raw power spectra and the three kinds of reduced feature vectors are shown in Tab.2. Obviously, the weighted feature reduction method has the advantage over the method of directly using FDR to reduce feature dimensionality of better recognition performance with the same computation complexity. The recognition rate of An-26 by reduced feature vectors based on Fisher's optimal projection subspace is lower than that by raw power spectra. The reason is that An-26 is a propeller-driven aircraft and its HRRPs which are modulated by propellers scatter widely in the feature space. It is difficult to seek an optimal projection subspace which can divide all features of Yark-42, Cessna citation S/II and An-26 well. The result is that the recognition performance on An-26 is not as good as on the other two, and the average recognition rate by the subspace method is lower than that by the weighted method. Therefore, the weighted feature reduction method proposed in this paper is more robust to different test data. Furthermore, the computation burden of the subspace method is much larger than that of the weighted method. According to Eq.(11), for an N -dimensional raw power spectrum (for our measured HRRP $N=128$), the computation of the subspace method is N times larger than that of the

weighted method.

Tab.2 A comparison in recognition performance between the feature reduction methods

Features	Recognition rates (%)			Average recognition rates (%)
	Yark-42	Cessna citation S/II	An-26	
Raw power spectra	94.25	83.25	71.75	83.08
Reduced feature vectors based on FDR	92.50	78.75	72.00	81.08
Weighted feature vectors based on FDR	94.50	89.75	82.00	88.75
Reduced feature vectors based on Fisher's optimal projection subspace	96.00	92.00	70.75	86.25

V. Conclusions

A new feature reduction method, the weighted feature reduction method based on FDR is presented in this paper. Compared with the method of using raw power spectra and some existing feature reduction methods, the weighted feature reduction method can not only reduce the feature dimensionality, but also improve the recognition performance with low computation complexity. In the recognition experiments based on measured data, the proposed method is robust to different test data and achieves good recognition results.

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